Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Proctored Exam — Formula Sheet

MCB80.1x · MCB80.2x · MCB80.3x · 37 °C · exam calculations only

Symbols: V_m membrane potential; E_{ion} reversal potential; g conductance (S); outward current positive unless noted. Pair with the **exam study guide** (concepts) and the **practice app** (timed MCQs).

EXAM CONSTANTS

Constant	≈ Value	Exam use
RT/F (37 °C)	26.7 mV	Nernst with ln
61.54 mV / decade	$\log_{10}$ Nernst	10× [out]/[in] → 61.5/z mV shift
2× [out] (z = 1)	+18.5 mV	E_{ion} more positive
4× [out] (z = 1)	+37 mV	fourfold K_{out} etc.
4× [out] (z = 2, Ca^{2+})	+18.5 mV	half slope

Core equations (Part 1)

Nernst

$E_{ion} = (61.54\ mV / z) \cdot \log_{10}([ion]_{out} / [ion]_{in})$

Goldman (K^+ + Na^+, ignore Cl^-)

$V_m = (RT/F) \cdot \ln((P_K[K^+]_o + P_{Na}[Na^+]_o) / (P_K[K^+]_i + P_{Na}[Na^+]_i))$

Ionic current

$I = g \cdot (V_m - E_{ion})$ · $\tau = R_m C_m$ · $\lambda = \sqrt{(R_m/R_i)}$

Passive steady state

$V_\infty = I_{inject} \cdot R_{in}$ (if R_m doubles, V_∞ doubles for fixed I)

SIGN CHECKS

- ↑ K_{out} → E_K less negative → V_m depolarizes.
- ↑ Na_{out} → E_{Na} more positive.
- $V_m = E_{ion}$ → zero driving force for that ion.
- Na^+/K^+ pump: 3 Na out, 2 K in (electrogenic).

Quick Part 2 · minimal

- Evoked PSP ≈ **n · q** (quanta × mini size).
- EPSPs/IPSPs sum within **τ** (temporal) and at soma (spatial).
- **NMDA LTP:** glutamate + depolarization → Mg^{2+} unblock → Ca^{2+} in.
- Shunt inhibition: ↑ g → ↓ R_{in} → smaller EPSPs.

UNITS (COMMON TRAP)

- 1 nS × 1 mV = 1 pA | 50 nS × (−70 mV) = **−3.5 nA**
- $I_C = C_m \cdot dV_m/dt$ at step onset

Worked examples (show your steps on the exam)

Example A — Nernst + Goldman resting V_m

Given at 37 °C: $[K^+]_i$ = 140, $[K^+]_o$ = 5, $[Na^+]_i$ = 15, $[Na^+]_o$ = 145 mM; $P_K : P_{Na}$ = 1 : 0.05; ignore Cl^-.

1. $E_K = 61.54 \cdot \log_{10}(5/140) \approx$ **−89.1 mV**
2. $E_{Na} = 61.54 \cdot \log_{10}(145/15) \approx$ **+60.6 mV**
3. Goldman: num = 1·5 + 0.05·145 = 12.25; den = 1·140 + 0.05·15 = 140.75
4. $V_m = 61.54 \cdot \log_{10}(12.25/140.75) \approx$ **−65.3 mV** (nearer E_K than E_{Na})

Do not plug "average of E_K and E_{Na}" unless the question gives only permeability ratios of the E values — use concentrations in Goldman.

Example B — Current from conductance

g_{Na} = 50 nS, V_m = −10 mV, E_{Na} = +60 mV (outward positive convention).

1. Driving force: $V_m - E_{Na} = -10 - 60 =$ **−70 mV**
2. $I = 50 \times 10^{-9}$ S $\times (-70 \times 10^{-3})$ V = $\mathbf{-3.5 \times 10^{-9}}$ **A = −3.5 nA** (inward Na^+)

Example C — Concentration fold-change (no calculator friendly)

Extracellular Na^+ doubled, intracellular unchanged → E_{Na} shifts by $61.54 \cdot \log_{10}(2) \approx$ **+18.5 mV** (more positive). Fourfold extracellular $K^+ \rightarrow E_K$ shifts ≈ **+37 mV** (less negative).